SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1710**

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

Total Marks:40

QUESTIONS: 4

Annexure A

CASE STUDY

Code of Conduct:

I Ravi Teja (192565088) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

23 + 23 + 18 + 18 + 9

= 91

Case Study Co3 AT1

Q1. Cybersecurity detection System:

The given facts are failed Login attempts, unusuall address and high data Transfer = True.

a) Propositional Logic.

F = Failed login Attempts.
U = Unusual IP Address.
O = odd login Time
H = High data Transfer
B = Brute Force Attack.
A = Account Compromise.
D = Data exfiltration.

$F \wedge U \rightarrow B$

$F \wedge O \rightarrow A$

$U \wedge H \rightarrow D$

F = True, U = True, H = True.

b) Forward Chaining.

Step 1:

$F \wedge U \longrightarrow B$

Since F = True and U = True,

$\therefore$, B = True.

→ Possible Brute Force Attack

Step 2:

using Rule 2:

$F \wedge O \rightarrow A$

O is not given as True.

$\therefore$, A Cannot be derived.

Step 3:

using Rule 3:

$U \wedge H \rightarrow D$

Since U = True & H = True,

$\therefore$, D = True.

→ Possible data exfiltration.

Brute force Attack = True.

Data exfiltration = True.

c, Backward chaining.

Goal: Data exfiltration.

find a rule whose conclusion is data exfiltration.

$D \leftarrow U \wedge H$

Check Subgoal.

U = True

H = True

Both Subgoal are satisfied by the factor

$\therefore$, $U \wedge H \rightarrow D$

Hence,

Session x indi

Thus,

Hence,
Session x indicates data exfiltration.
Thus, backward chaining successfully proves the goal.

Q2: Airport priority Landing System.

a) $\forall x$ [AirCraft (x) $\wedge$ Emergency declared (x) $\Rightarrow$ Priority Clearance (x)].

Given facts:
. AirCraft (flight 2)
. Emergence declared (Flight 2)

Match:
AirCraft (x) = AirCraft (Flight 2)

$\therefore$,
$\theta = \{x / flight\ 2\}$

after Substitution
AirCraft (Flight 2) $\wedge$ emergency Declared (Flight 2) $\Rightarrow$
Priority clearance (Flight 2)

using Rule 2:
Priority clearance (x) $\rightarrow$ Runway Reserved (x)

Substitution
$\theta = \{x / Flight\ 2\}$

$\therefore$, Runway Reserved (Flight 2)

c) Two emergencies:

The current KB can identify emergency aircraft but does not define conflict resolution for two simultaneous emergencies.

Additional rules for priority and runway allocation are required.

Q3: Industrial Equipment maintenance.

Rules:

$M \rightarrow N$

$N \land H \rightarrow S$

$\neg S \land H \rightarrow R$

fact

$M = True, H = True$.

a) CNF

P1:

$M \rightarrow N$

$\neg M \lor N$

P2:

$N \land H \rightarrow S$

$\neg N \lor \neg H \lor S$

P3:

2. emergency Declared (Flight2)
3. Air Craft (FlightY)
4. QueuedAfter (FlightY, flight2).

Rule 3 with:

X = Flight Y
Y = Flight 2

gives : AirCraft (flight Y) ∧ Queued After (flightY, flight2)
∧ Runway Reserved (Flight2).
All Three are True.

∴ Negate the goal:
¬Delay (FlightY)

Rule 3 gives:
¬AirCraft (FlightY) ∨
¬QueuedAfter (FlightY, flight2) ∨
¬Runway reserved (Flight2) ∨ Delay (FlightY).

using the three True fact, this reduced to
Delay (FlightY).

Resolving With ¬Delay (FlightY) given
∴, Delay (FlightY) is proved.

. N Can't be derived.
. S Can't be derived.
. R Can't be derived.

∴, Production Risk doesnot follow.

Q4: Minefield Robot:

Vibration [2,1], Smoke [1,1], Beacon [2,2]

a) Belief state

Rules:

Vibration [2,1] → mine adjacent
Smoke [1,1] → fire adjacent.
Beacon [2,2] → safe Zone Marker [2,2]

∴, mine is possible near [2,1]
Fire is possible near [1,1]
Safe-Zone marker is at [2,2].

b) Forward chaining:

Plain text

Vibration [2,1]
↓
mine adjacent To [2,1]
Smoke [1,1]

$\neg S \land H \rightarrow R$

$S \lor \neg H \lor R$

facts:

M, H.

b) Resolution refutation

Goal:

S

Negate:

$\neg S$

From:

$\neg M \lor N, M$

derive:

N

Then:

$\neg N \lor \neg H \lor S, N$

gives:

$\neg H \lor S$

using H:

$\neg H \lor S, H$

gives: S

Finally

$S, \neg S \rightarrow \square$

$\therefore$, True.

c) Remove machine overheating by Without M:

↓

Fire adjacent To [1,1]

Beacon [2,2]

↓

Safe Zone Marker [2,2]

Thus the possible hazard are near [2,1] & [1,1]

c) Path safety:

Path

[1,1] → [2,1] → [2,2]

- [1,1] → possible fire nearby.
- [2,1] → possible mine nearby.
- [2,2] → ~~Safe~~ Zone marker.

Since KB does not prove the first Two cells are Safe:

Path is NoT guaranteed Safe.